

Mayank Keoliya

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EDUCATION

University of Pennsylvania

Aug. 2023 – Present

PhD, Computer and Information Science

Advisor: Mayur Naik

National University of Singapore

Aug. 2019 – May 2023

Bachelor's in Computer Science

Advisor: Ilya Sergey

RESEARCH INTERESTS

I aim to enhance the safety and adoption of ML models in healthcare by (1) developing new interpretable models for critical applications like cancer treatment prediction and sepsis detection, (2) enabling multi-modal reasoning over diverse clinical data sources, and (3) ensuring these models are explainable and trusted by clinicians in high-stakes decisions.

PUBLICATIONS

 Distinguished/Spotlight Paper

LASER: A Neuro-Symbolic Framework for Learning Spatial-Temporal Scene Graphs with Weak Supervision

Jiani Huang, Ziyang Liu, **Mayank Keoliya**, Mayur Naik, Sernam Lim

Under submission at ICLR '25

Synthesizing Faithful Explanations for Treatment Effect Estimation

Yinjun Wu*, **Mayank Keoliya***, Kan Chen, Neelay Velingker, Emily Getzen, Qi Long, Mayur Naik, Ravi Parikh

ICML '24

Mostly Automated Proof Repair for Verified Libraries

Kiran Gopinathan, **Mayank Keoliya**, Ilya Sergey

PLDI '23

Simulating Noisy Channels for DNA Storage

Mayank Keoliya, Puru Sharma, and Djordje Jevdjic

ISPASS '22

ONGOING PROJECTS

Trustworthy AI for Early Sepsis Detection and Breast Cancer Treatment

Lead PhD student on a ARPA-H grant to develop trustworthy AI for early sepsis prediction, and treatment recommendation for breast cancer. Working with clinicians at Penn to train and deploy explainable yet accurate models on data from 300K patients across Pennsylvania.

DISCRET for Mortality Prediction

Applying our prior work on DISCRET to mortality prediction. Our model generates explanations, which are quantitatively more robust and faithful than traditional post-hoc methods like LIME and SHAP. Currently in user studies with over 50 oncologists at Penn Medicine and other health systems.

WORK EXPERIENCE

VERSE Lab, NUS

Research Assistant

May 2022 – April 2023

- Part of team that devised a novel proof repair technique which enables dynamic certification of critical systems such as financial trading and aircraft navigation
- Designed and implemented an efficient invariant synthesis algorithm
- Conducted comprehensive evaluation of our work on a diverse set of programs

National University of Singapore

Research Assistant

Aug 2021 – May 2022

- Investigated noise in DNA sequencing, particularly with next-gen Nanopore sequencing
- Discovered key insights about the relationship between sequencing errors and molecule position within a DNA strand
- Work was published as a poster at ISPASS'22

Garena

Summer Intern

May 2021 – Aug 2021

- Developed an alerts management system which enables teams to triage and fix errors across all of Garena's games
- Scaled the triage system to handle alerts from over 2.4 billion requests per day

AWARDS

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| • Outstanding Undergraduate Researcher , NUS (top 5 in a cohort of 6k) | 2023 |
| • GIC Medal , NUS Honors College (best computing student) | 2023 |
| • President's Honour Roll , NUS Honors College | 2023 |
| • SIGPLAN-M Travel Grant | 2022 |

TEACHING EXPERIENCE

TA, Advanced Functional Programming, CIS 552

University of Pennsylvania

Aug 2024 – Dec. 2024

TA, Intro to Programming, CS1101S

National University of Singapore

Aug 2020 – Dec. 2020

REFERENCES

Mayur Naik (PhD Advisor)
Professor and Graduate Chair
Computer and Information Science
University of Pennsylvania
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☎ 215-573-1856

Ravi B Parikh (Collaborator)
Associate Professor
School of Medicine
Emory University
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